

Student Name: Muhammad Hamzah Iqbal Student ID: F24604018	Submission Date: March 24,2025
--	---------------------------------------

Computer Programming (CEN1008)

Assignment # 1

(CLO # 1 → PLO 1)

Total Marks: 20

Q1. List down the important features of Python language and explain them briefly. (10 points)

Python is a high-level, adaptable, and easy-to-learn programming language widely used in web development, data science, automation, artificial intelligence, and more. It is known for its simple syntax, extensive libraries, and strong community support, making it an ideal choice for both beginners and experienced developers. Below are some of Python's key features:

1. Portable Language

Python is cross-platform, meaning it runs on Windows, macOS, and Linux without modification.

2. Standard Library

Python includes built-in modules for tasks like math, data handling, and file operations. It also supports powerful external libraries like NumPy, Pandas, and TensorFlow, making development faster.

3. High-Level Language

Python is easy to read and write, unlike low-level languages like C or C++, making it accessible to beginners.

4. Easy to Learn & Use

Python has a simple syntax, allowing beginners to write programs quickly. For example, adding two numbers takes just one line:

```
a = 8
```

```
b = 9
```

```
print(a + b)
```

In contrast, other languages require more lines of code.

5. Dynamic Language

You don't need to specify variable types. For example:

```
a = 7 # No need to declare 'int a = 7' like in C.
```

6. Extensible Language

Python code can be combined with C or C++, enhancing performance and functionality.

7. Interpreted Language

Python executes code line by line, making debugging easier since compilation isn't required.

8. Object-Oriented Language

Python supports OOP principles like abstraction, encapsulation, inheritance, and polymorphism, making it flexible for different programming styles.

9. Free & Open Source

Python is free to download and use, with a vast community contributing to its development.

10. GUI Support

Python offers multiple libraries for Graphical User Interfaces (GUIs), making it useful for software and web development.

Q2. Explain the importance of Anaconda framework/package and how it is useful for Python programmers. (10 points)

Importance of Anaconda framework/package:

- Includes Conda, which simplifies package installation and dependency management. Supports isolated environments, preventing conflicts between different Python projects.
- Comes with 1,500+ pre-installed Python libraries like NumPy, pandas, Matplotlib, and scikit-learn, saving setup time.
- Features Anaconda Navigator, a GUI for managing Python environments and packages without using the command line.
- Designed for Python-based data science, AI, and ML workflows, providing easy access to essential tools.
- Enables maintaining specific Python library versions, ensuring consistency in research and production projects.

How Anaconda is useful for Python Programmers:

- Anaconda lets programmers concentrate more on coding and less on configuration by offering a single platform for package management and environment setup.
- Anaconda is a flexible tool for data scientists who might deal with various programming languages because it supports R and other languages in addition to Python.
- Users may access a multitude of resources, tutorials, and help thanks to Anaconda's sizable community and comprehensive documentation.
- Anaconda's smooth integration with Jupyter Notebooks, a well-liked program for data analysis and visualization, enables interactive scripting and simple result sharing.
- By using conda environments, users can maintain specific versions of libraries for different projects, ensuring reproducibility of results. This is critical in research and production environments where consistent results are necessary.